POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

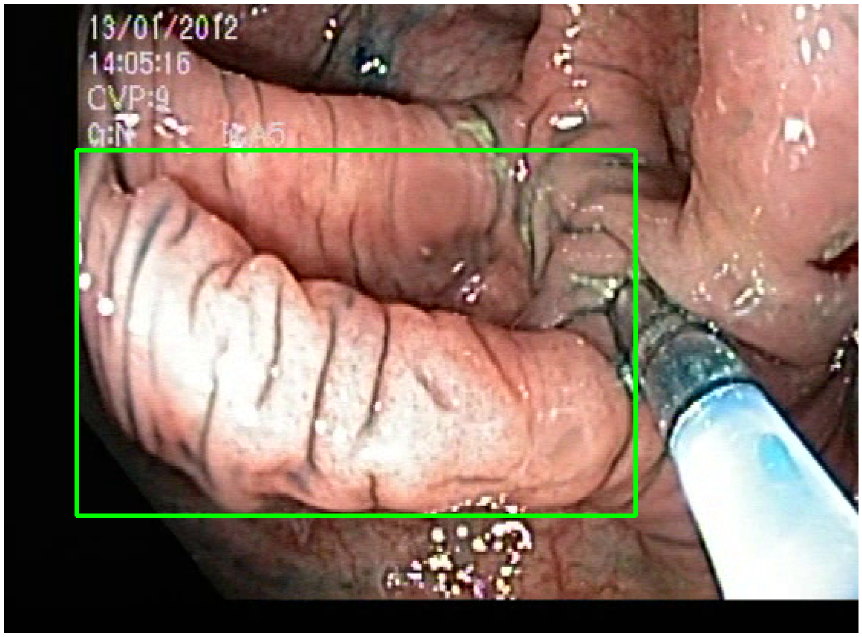
Video ID:	b71574a3-0fdd-4040-9fda-ca2b4a2634bf
Total Frames:	3781
Frame Rate:	25.0 fps (assumed)
Duration:	151.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	281	✓
RT-DETR Detections	539	✓
MedSAM2 Detections	698	✓
Consensus (3-Model Agreement)	7	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	7	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 75 frames)



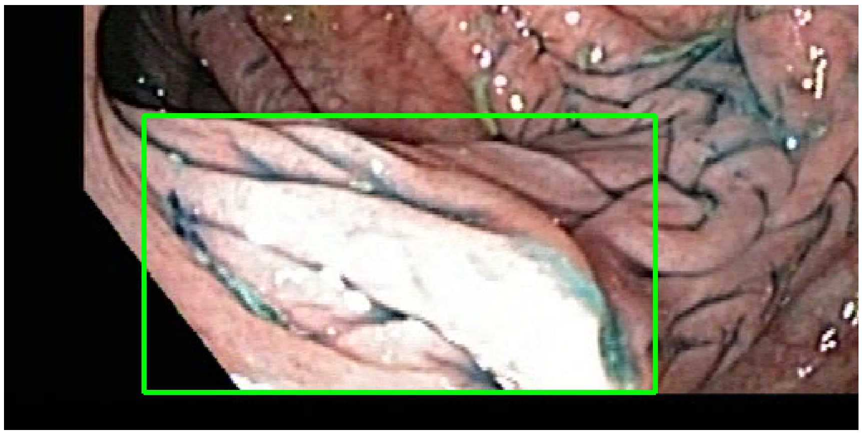
Feature	Value	Interpretation
Frame Index	3288	Frame 3288 of 3781
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**SERRATED_POLYP**	Type confidence: 83.1%
Redness Score	0.112	Color-based vascularization
Relative Radius	43.3%	Polyp size as % of frame diagonal
Texture Score	0.305	Surface morphology
Vessel Visibility	0.215	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified

Detection Confidence	76.6%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 78 frames)

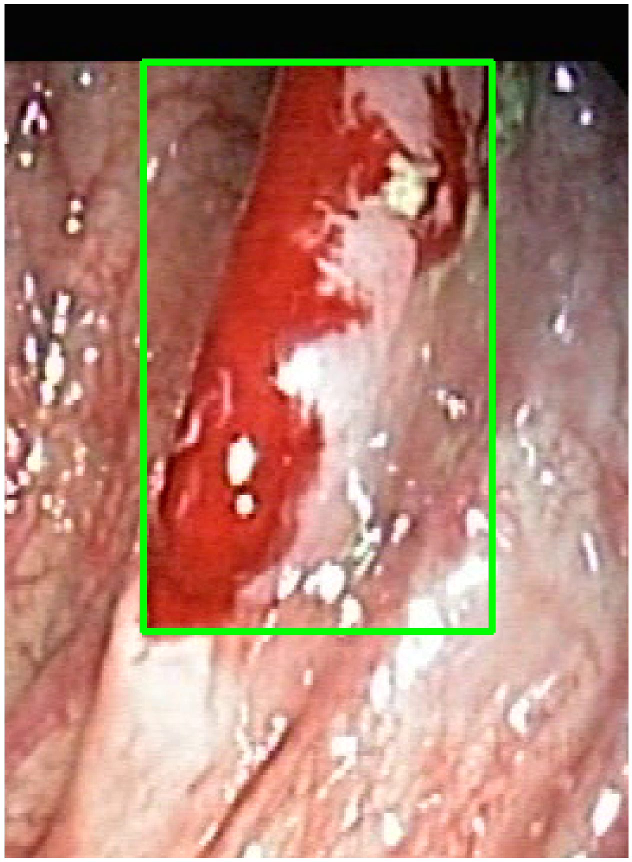


Feature	Value	Interpretation
Frame Index	1286	Frame 1286 of 3781
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 92.0%
Redness Score	0.067	Color-based vascularization
Relative Radius	14.1%	Polyp size as % of frame diagonal
Texture Score	1.000	Surface morphology
Vessel Visibility	0.001	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.3%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 62 frames)



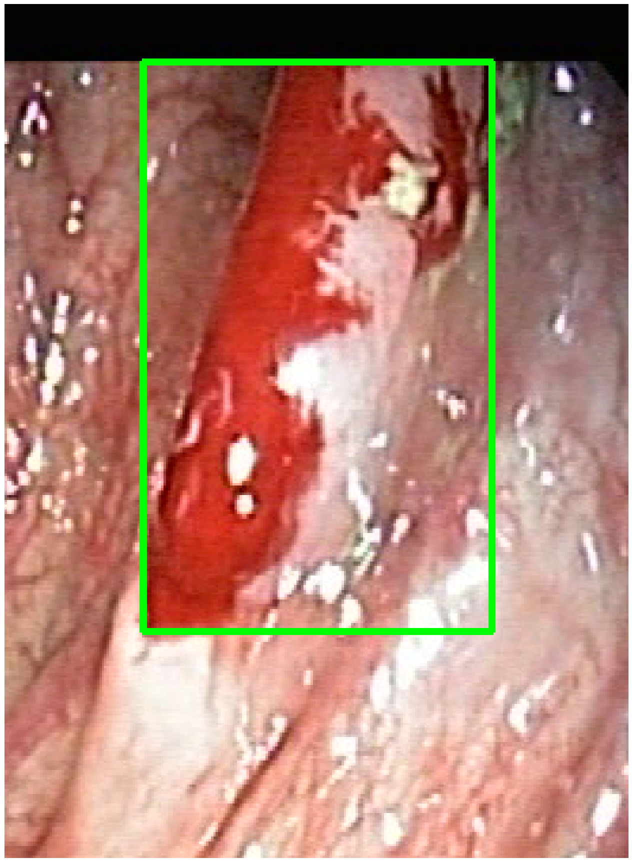
Feature	Value	Interpretation
Frame Index	2860	Frame 2860 of 3781
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLOv8 + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.216	Color-based vascularization
Relative Radius	34.1%	Polyp size as % of frame diagonal
Texture Score	0.206	Surface morphology
Vessel Visibility	0.734	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	75.2%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 62 frames)



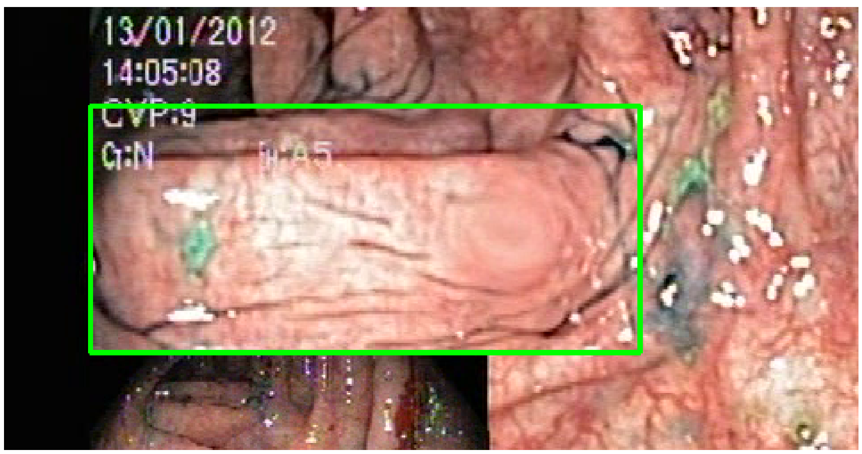
Feature	Value	Interpretation
Frame Index	2860	Frame 2860 of 3781
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.216	Color-based vascularization
Relative Radius	34.1%	Polyp size as % of frame diagonal
Texture Score	0.206	Surface morphology
Vessel Visibility	0.734	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	73.5%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #5 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 27 frames)



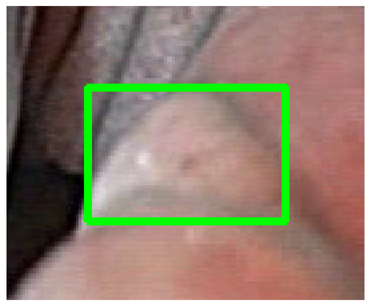
Feature	Value	Interpretation
Frame Index	3088	Frame 3088 of 3781
Detection Method	Detector Consensus (YOLO + RT-DETR agreement)	(MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	**SERRATED_POLYP**	Type confidence: 92.0%
Redness Score	0.111	Color-based vascularization
Relative Radius	36.3%	Polyp size as % of frame diagonal
Texture Score	0.557	Surface morphology
Vessel Visibility	0.165	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	62.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #6 — LIFTED_POLYP [LOW_RISK] (Tracked across 23 frames)



Feature	Value	Interpretation
Frame Index	1066	Frame 1066 of 3781
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 59.9%
Redness Score	0.062	Color-based vascularization
Relative Radius	8.2%	Polyp size as % of frame diagonal
Texture Score	0.107	Surface morphology
Vessel Visibility	0.012	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	70.8%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #7 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 21 frames)

Feature	Value	Interpretation
Frame Index	1185	Frame 1185 of 3781
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 unavailable)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 83.0%
Redness Score	0.091	Color-based vascularization
Relative Radius	23.0%	Polyp size as % of frame diagonal
Texture Score	0.556	Surface morphology
Vessel Visibility	0.032	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	69.3%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:
A total of 7 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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